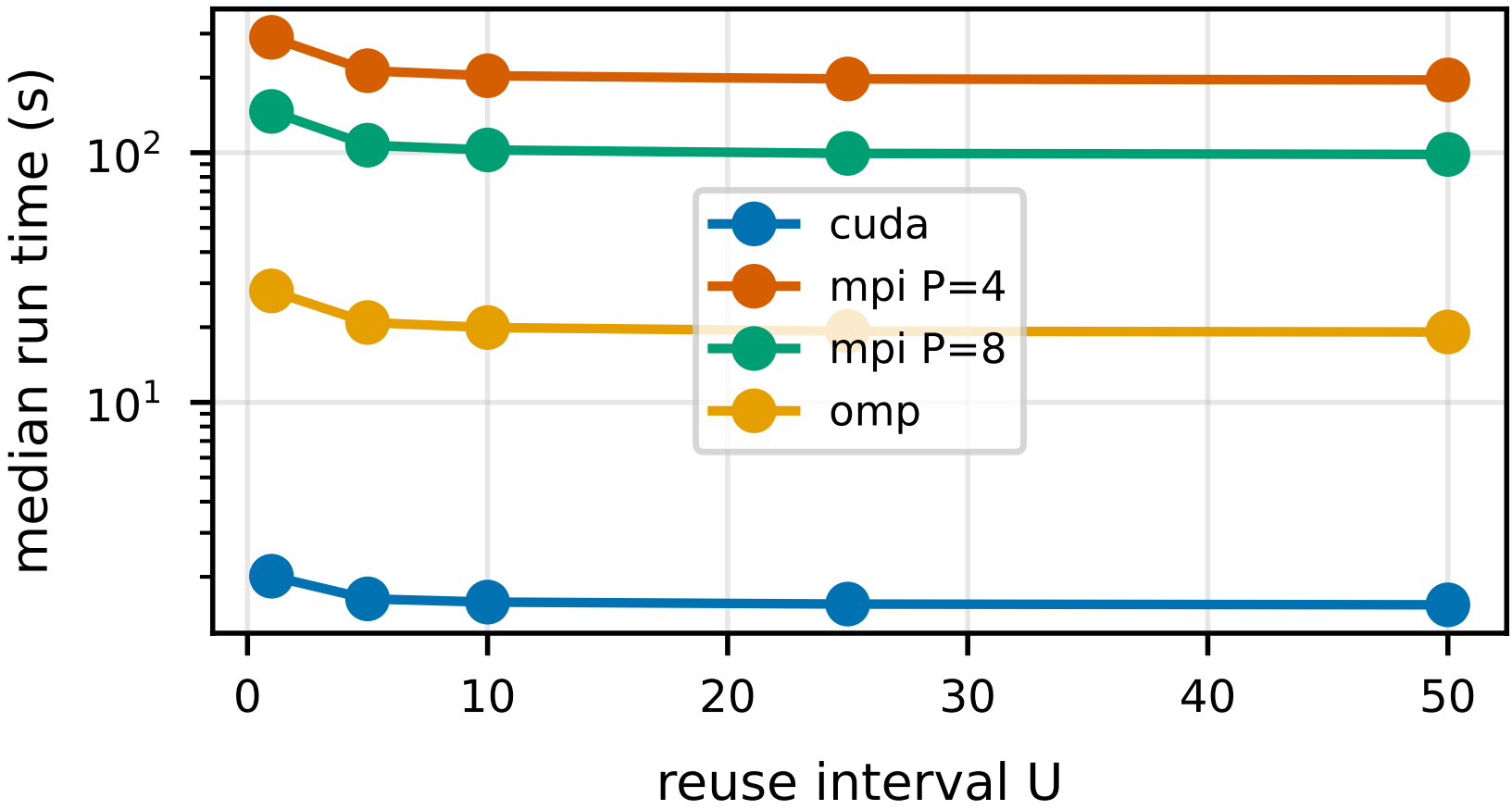
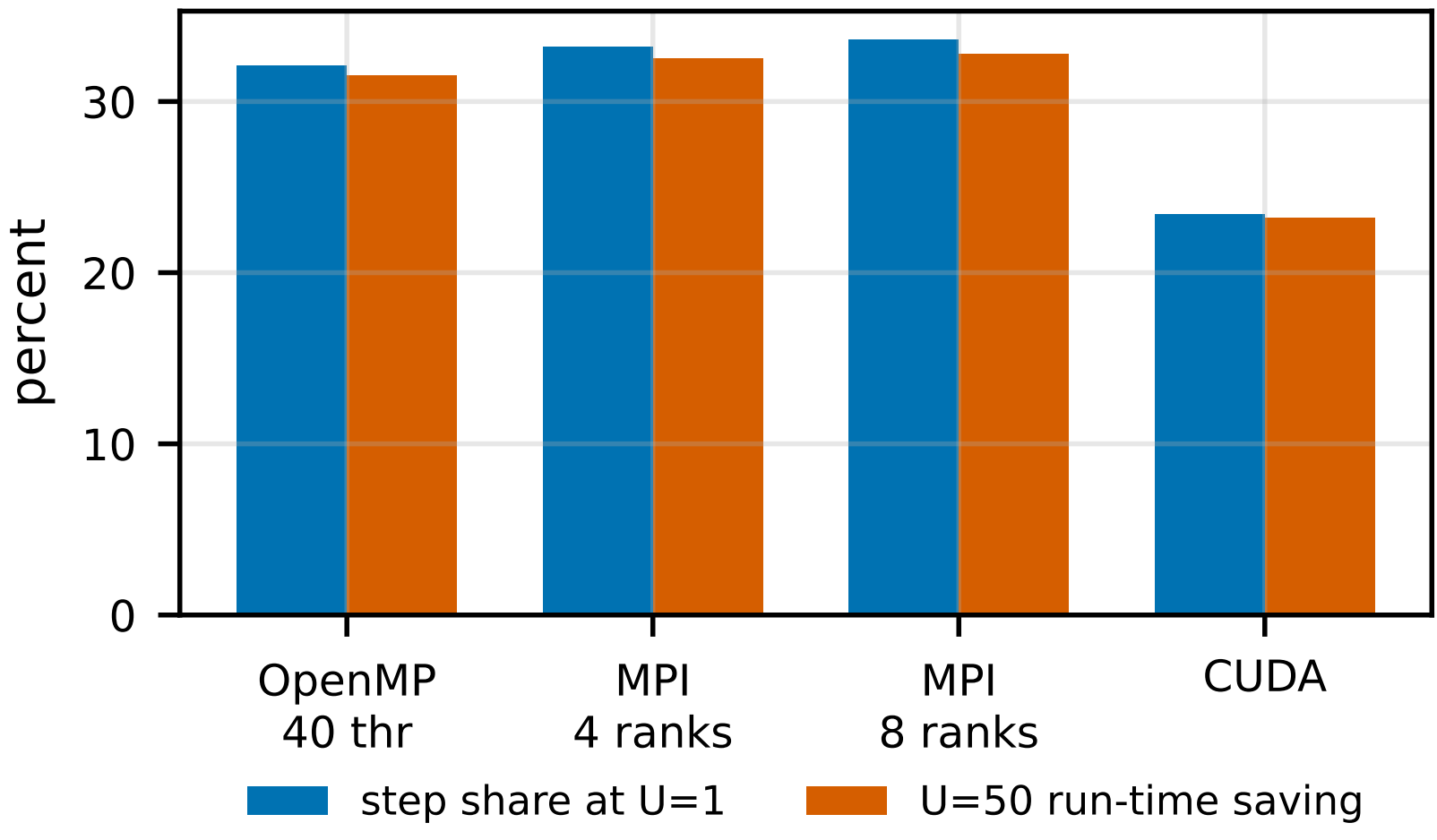

median run time (s)
10^2
10^1
cuda
mpi P=4
mpi P=8
omp
0
10
20
30
40
50
reuse interval U


percent
30
20
10
0
OpenMP
40 thr
MPI
4 ranks
MPI
8 ranks
CUDA
step share at U=1
U=50 run-time saving